

IoT-Based Smart Agriculture Monitoring System Using Soil Moisture and IR Sensors with MQTT and Grafana

PROJECT NO : 0105

SUBMITTED BY:

KAVIYA S

24UCS084

KAMARAJ COLLEGE OF ENGINEERING AND TECHNOLOGY

INTRODUCTION

Agriculture plays a vital role in food production, and efficient monitoring of field conditions is essential for improving crop yield. Traditional farming methods often depend on manual observation, which can be time-consuming and less accurate. The Internet of Things (IoT) enables smart farming by connecting sensors and devices to monitor agricultural parameters in real time. In this project, a Soil Moisture Sensor and an IR Sensor are used to collect field data, which is transmitted through the MQTT protocol. The collected data is stored in a database and visualized using Grafana dashboards, allowing farmers to monitor agricultural conditions remotely and make informed decisions for better crop management.

PROBLEM STATEMENT

Agricultural fields are often monitored manually, making it difficult to obtain accurate and real-time information about soil and crop conditions. Improper irrigation due to a lack of soil moisture monitoring can lead to water wastage and reduced crop productivity. Additionally, farmers may not have an efficient method to continuously observe field activities and environmental changes. Therefore, there is a need for an IoT-based smart agriculture system that can automatically collect sensor data, transmit it in real time, store it for analysis, and provide easy visualization through dashboards to support better farming decisions.

OBJECTIVES

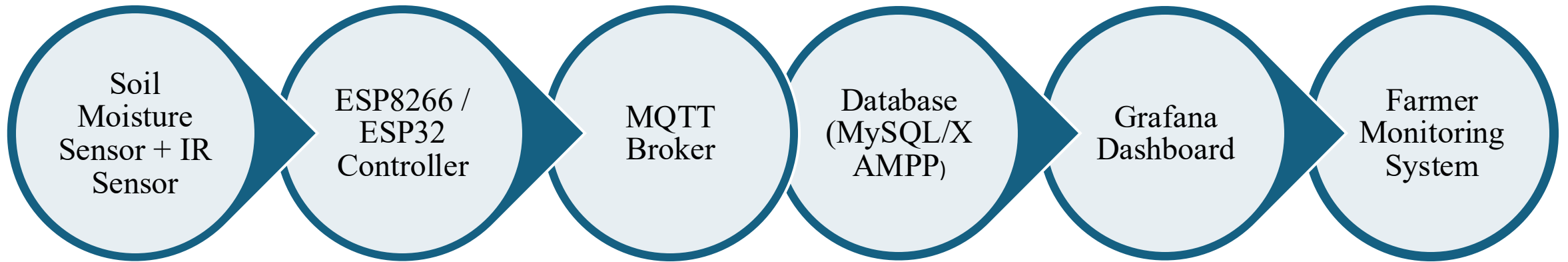
- ❑ Monitor soil moisture levels continuously.
- ❑ Detect crop or object presence using IR sensors.
- ❑ Transmit sensor data using MQTT protocol.
- ❑ Store sensor readings in a database.
- ❑ Visualize real-time and historical data using Grafana dashboards.



PROPOSED SOLUTION

- ❖ Soil Moisture Sensor measures soil water content.
- ❖ IR Sensor detects movement or object presence.
- ❖ ESP8266/ESP32 collects sensor readings.
- ❖ MQTT Broker receives and distributes sensor data.
- ❖ Database stores collected data.
- ❖ Grafana dashboard displays live monitoring results.

SYSTEM ARCHITECTURE



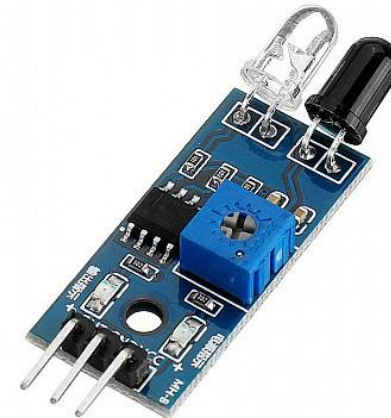
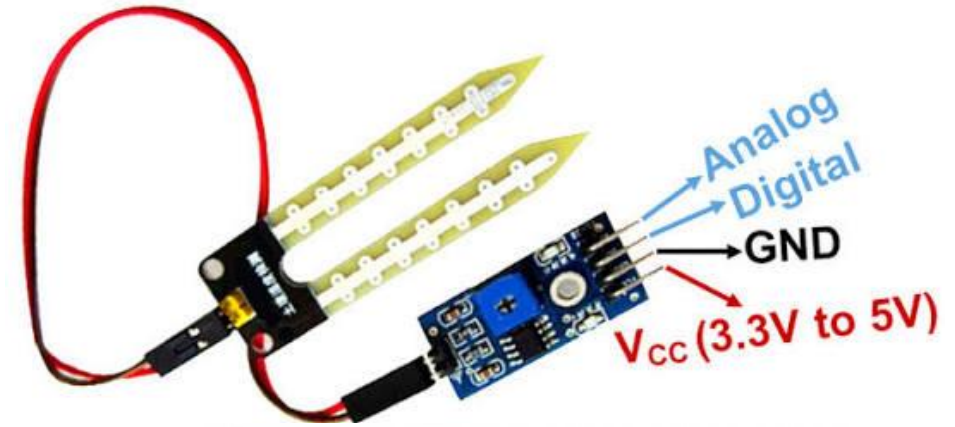
REQUIREMENTS

Hardware

- ESP8266 (ESP-12E) / ESP32
- Soil Moisture Sensor
- IR Sensor
- Breadboard
- Jumper Wires

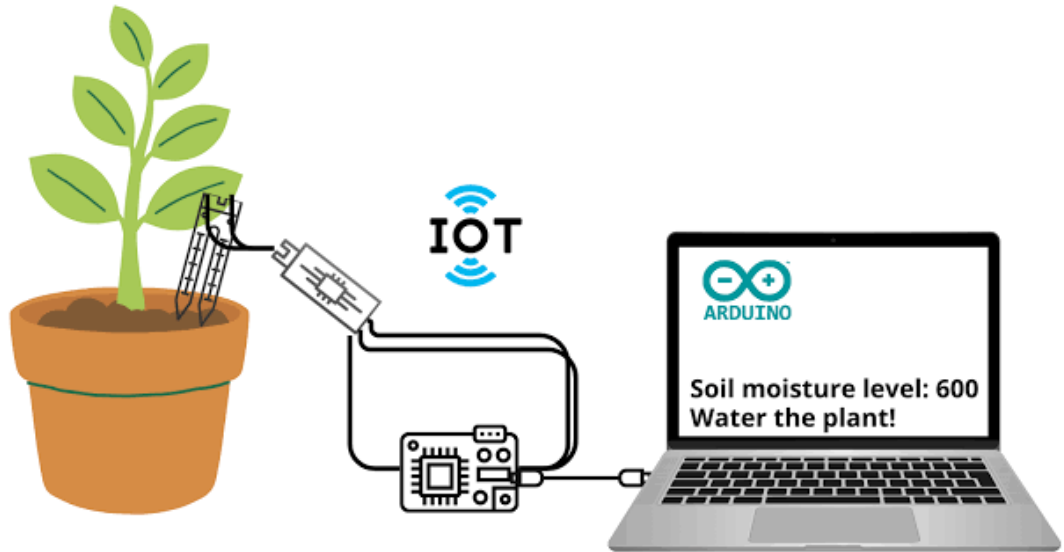
Software

- Arduino IDE
- MQTT Broker (EMQX/Mosquitto)
- XAMPP (MySQL Database)
- Grafana
- Wi-Fi Network



METHODOLOGY

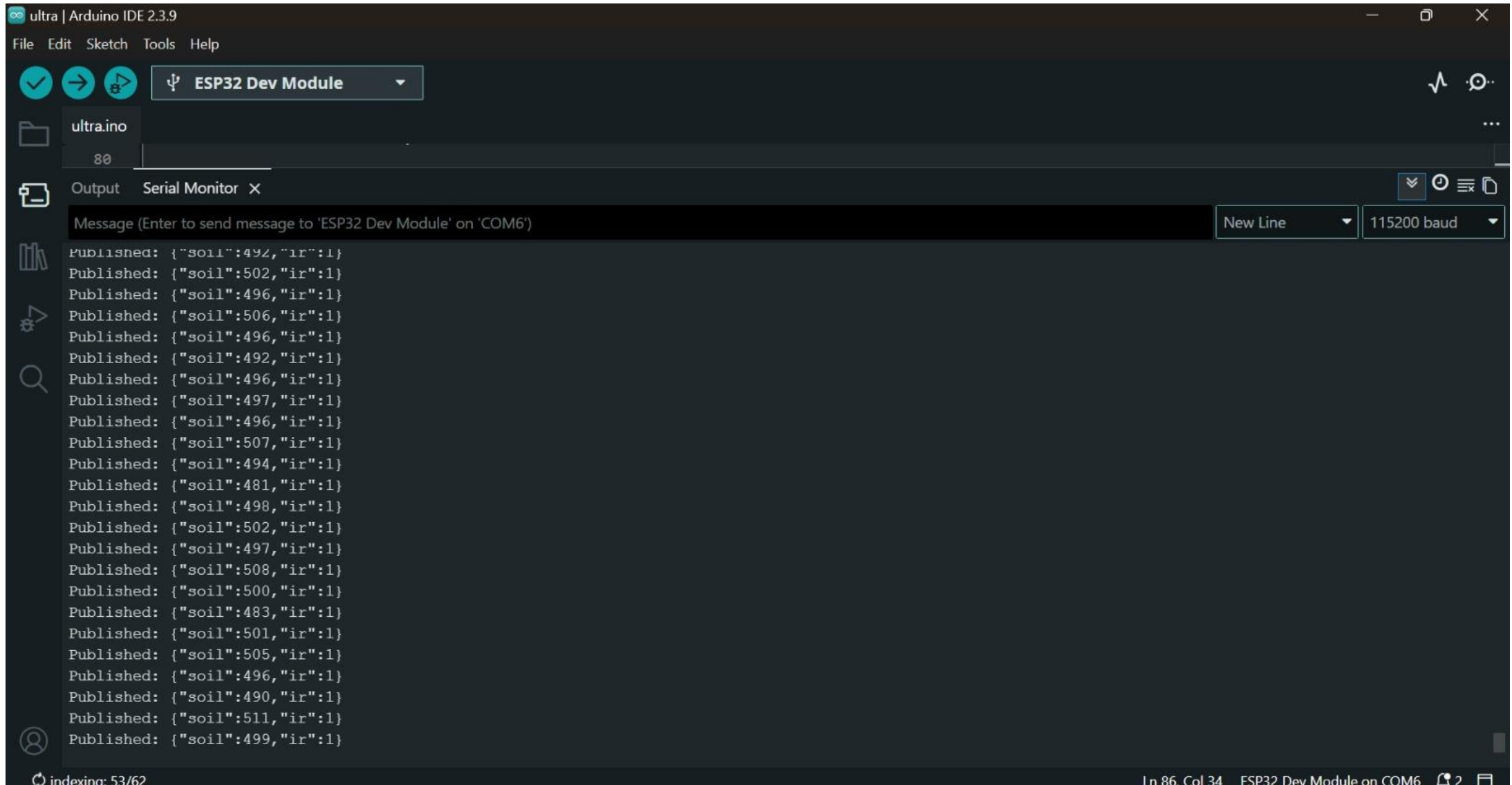
- ✓ Connect Soil Moisture and IR Sensors to
- ✓ Read sensor values continuously.
- ✓ Connect ESP8266 to Wi-Fi.
- ✓ Publish sensor data to MQTT Broker.
- ✓ Store received data in MySQL Database
- ✓ Retrieve data from database.
- ✓ Display data using Grafana Dashboard.
- ✓ Monitor agricultural conditions in real time.



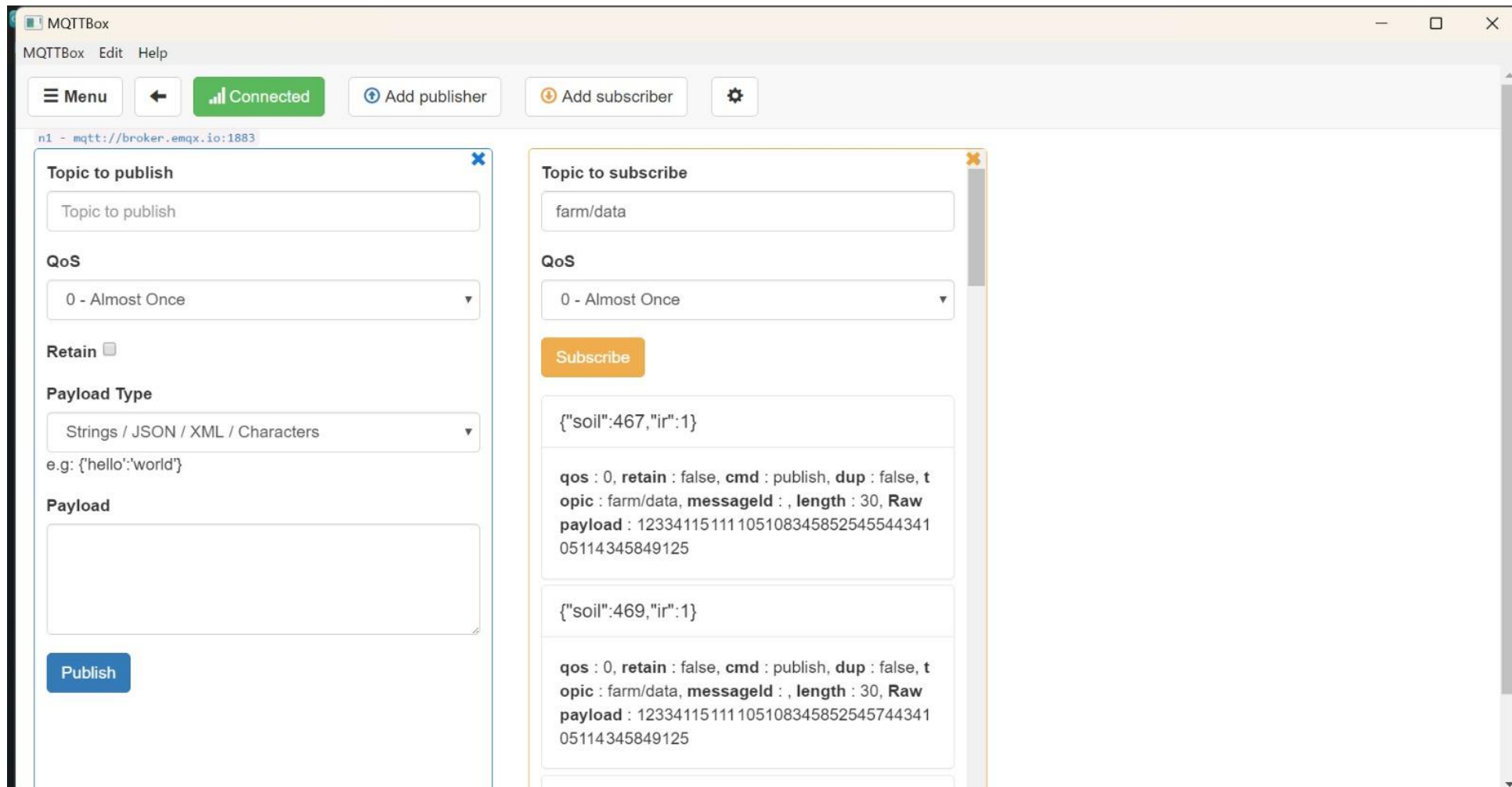
RESULT

The implemented IoT-based Smart Agriculture Monitoring System successfully collected soil moisture and IR sensor data and transmitted it through the MQTT protocol. The sensor readings were stored in the database and visualized using Grafana dashboards for real-time monitoring and historical analysis. The system provided accurate and continuous field information, enabling efficient monitoring of soil conditions and field activities. The results demonstrate that the proposed solution can support smart farming practices by improving resource management and reducing the need for manual observation.

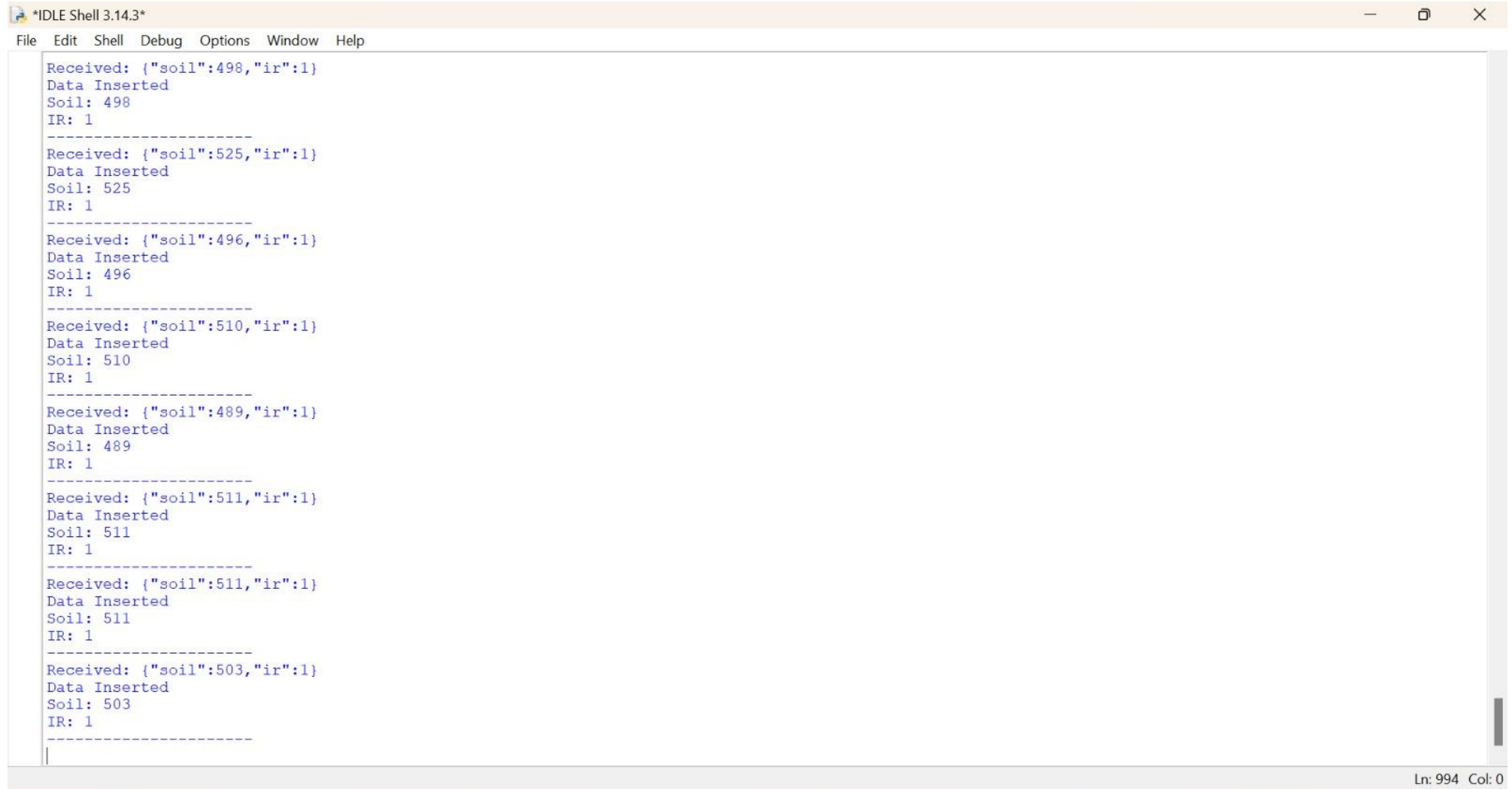
Arduino Serial Monitor Screenshot



Mqtt Box data Receiving Screenshot



Python Receiver code Screenshot

A screenshot of the IDLE Shell 3.14.3* window. The window has a menu bar with 'File', 'Edit', 'Shell', 'Debug', 'Options', 'Window', and 'Help'. The main text area displays the output of a Python receiver code. The output consists of several blocks of text, each separated by a dashed line. Each block starts with 'Received: {"soil":..., "ir":1}', followed by 'Data Inserted', 'Soil: ...', and 'IR: 1'. The 'Soil' values are 498, 525, 496, 510, 489, 511, 511, and 503. The status bar at the bottom right shows 'Ln: 994 Col: 0'.

```
*IDLE Shell 3.14.3*
File Edit Shell Debug Options Window Help

Received: {"soil":498,"ir":1}
Data Inserted
Soil: 498
IR: 1
-----
Received: {"soil":525,"ir":1}
Data Inserted
Soil: 525
IR: 1
-----
Received: {"soil":496,"ir":1}
Data Inserted
Soil: 496
IR: 1
-----
Received: {"soil":510,"ir":1}
Data Inserted
Soil: 510
IR: 1
-----
Received: {"soil":489,"ir":1}
Data Inserted
Soil: 489
IR: 1
-----
Received: {"soil":511,"ir":1}
Data Inserted
Soil: 511
IR: 1
-----
Received: {"soil":511,"ir":1}
Data Inserted
Soil: 511
IR: 1
-----
Received: {"soil":503,"ir":1}
Data Inserted
Soil: 503
IR: 1
-----
|

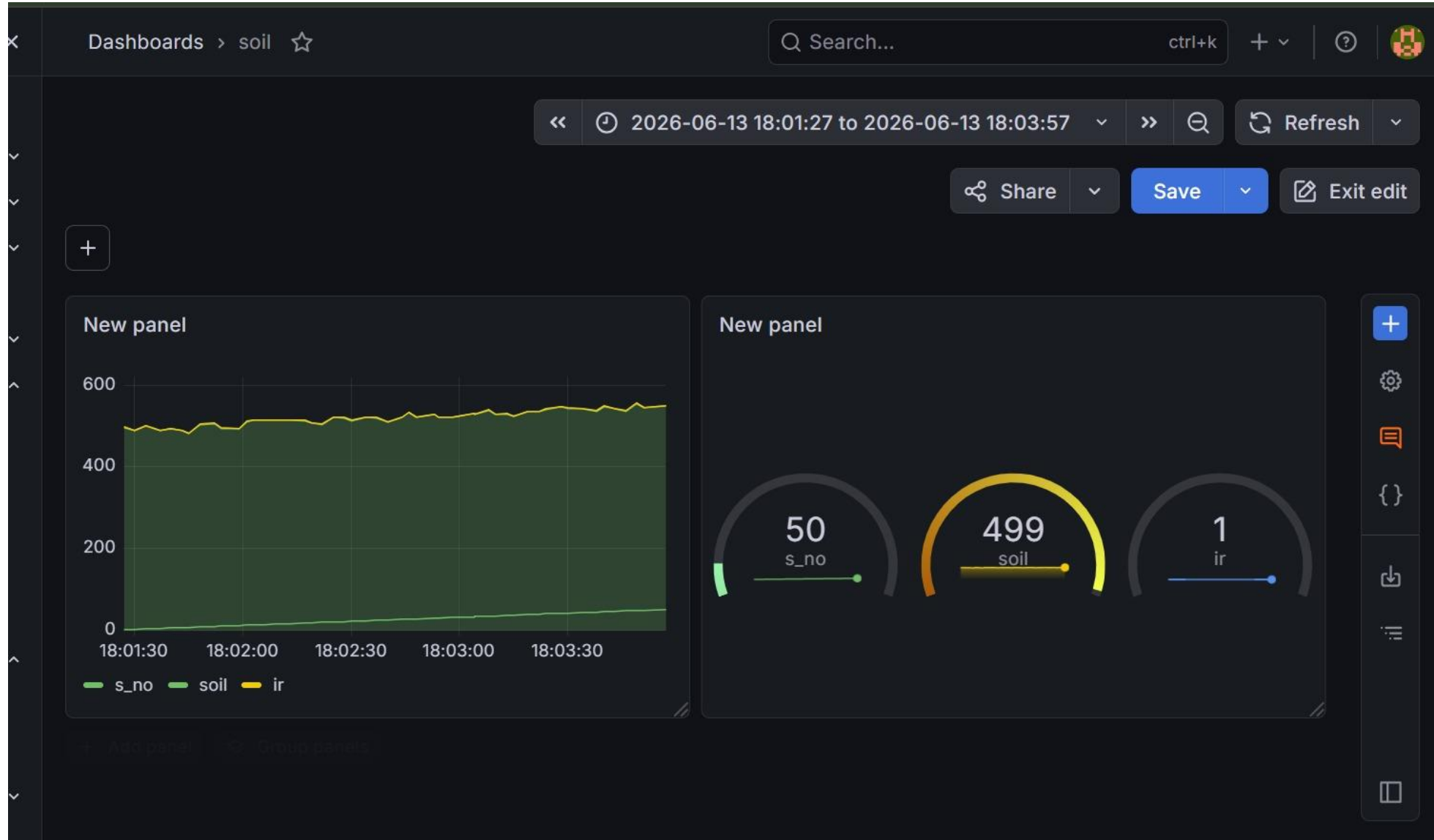
Ln: 994 Col: 0
```

Database Datastorage Screenshot

The screenshot displays the phpMyAdmin web interface. The left sidebar shows a tree view of databases, with 'smart_agriculture' selected and its 'sensor_data' table highlighted. The main panel shows the 'sensor_data' table structure and data. The table has four columns: 's_no', 'soil', 'ir', and 'timestamp'. There are 17 rows of data, each with a checkbox for selection and links for 'Edit', 'Copy', and 'Delete'.

	s_no	soil	ir	timestamp
☐	1	496	1	2026-06-13 12:31:27
☐	2	487	1	2026-06-13 12:31:30
☐	3	498	1	2026-06-13 12:31:33
☐	4	485	1	2026-06-13 12:31:37
☐	5	489	1	2026-06-13 12:31:40
☐	6	483	1	2026-06-13 12:31:43
☐	7	475	1	2026-06-13 12:31:45
☐	8	496	1	2026-06-13 12:31:48
☐	9	497	1	2026-06-13 12:31:52
☐	10	485	1	2026-06-13 12:31:54
☐	11	483	1	2026-06-13 12:31:59
☐	12	499	1	2026-06-13 12:32:01
☐	13	502	1	2026-06-13 12:32:03
☐	14	501	1	2026-06-13 12:32:06
☐	15	499	1	2026-06-13 12:32:09
☐	16	498	1	2026-06-13 12:32:13
☐	17	496	1	2026-06-13 12:32:17

Grafana Screenshot



CONCLUSION

The proposed IoT-based Smart Agriculture Monitoring System successfully monitors soil moisture and IR sensor data in real time. MQTT ensures reliable data transmission, while MySQL stores sensor readings efficiently. Grafana provides easy-to-understand dashboards for farmers. The system helps optimize irrigation, improve crop management, and supports smart farming practices.